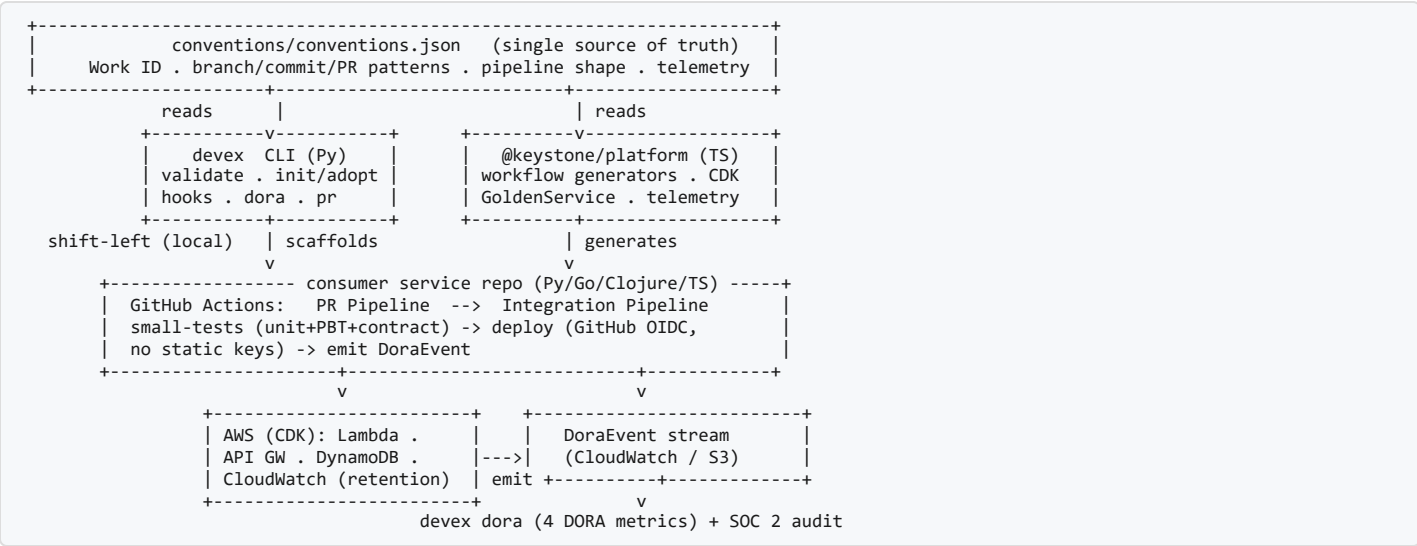


Keystone — Architecture & Platform Strategy

A Golden Path for 10+ independent, full-cycle engineering teams. One thesis runs through every decision below: **we don't standardize the *metric* — we standardize the *source of the metric*.** Comparability, governance, and audit become structural properties of the platform, not conventions teams are asked to remember.

Architecture



The CLI (developer-side) and the framework (platform-side) **never import each other**; they integrate only through `conventions.json` and the documented `DoraEvent` schema. That is what lets each be versioned and Git-installed independently, and what guarantees local validation and CI enforcement can never drift.

Homologation — how 10+ teams adopt consistently

- Convention over configuration.** The golden path is the *default* that `devex init` (new service) and `devex adopt` (existing service) generate: PR template, CI workflows, hooks, and a `keystone.json`. Adoption is one command, not a migration project — if following the standard is harder than bypassing it, that is a platform bug.
- One source, two consumers.** Every convention (Work ID `FIN-123`, branch/commit/PR patterns, pipeline shape, telemetry schema) is defined **once** in `conventions.json`. The CLI bundles a synced copy (CI fails on drift); the framework reads it to generate workflows. No pattern is hard-coded in Python or TypeScript.
- Identical pipelines by construction.** Because the framework — not the application — generates CI and emits telemetry, every team's `small-tests`, `deploy` steps, and `DoraEvent` are identical regardless of language.

Scalability — the platform team is not a bottleneck

- Inner-source by design.** Rough edges become small reviewed PRs that improve the platform for the *next* team. Dogfooding the golden path on a real service (Transactionify) surfaced **six platform gaps**, each fixed as a merged inner-source PR — the platform got better *because* it was used.
- Self-serve extension.** Adding a language is a contracts-only change behind one interface (`LANGUAGE_TOOLCHAINS`), documented as a `/new-language` recipe — no platform-team gatekeeping for the common case.
- Independent, registry-less distribution.** Two SemVer-tagged packages install straight from Git (`uv tool install ...#subdirectory=`, `pnpm add ...#path:`). The framework ships a **prebuilt dist/** (ADR-0004) so installs are zero-config on any package manager.

Shift-left — validation closer to the developer

- Same rules, three places.** `devex standards-check` and the installed git hooks run the **exact** `conventions.json` rules the generated CI re-runs — a failure surfaces on the workstation, not on a PR an hour later.
- Fail fast, act locally.** `devex pipeline run --local` simulates the pipeline before a push; the API-contract check validates the OpenAPI schema **pre-deploy** (server-less) and fuzzes the **deployed** URL post-deploy — a split that only emerged from a real adoption.

Distribution, security & cost

GitHub **OIDC** for AWS — there is no long-lived key to leak. CDK constructs are **cdk-nag-clean** with enforced CloudWatch log retention, `project=keystone` tags, and `RemovalPolicy.DESTROY`; AWS Budgets alarms guard the trial account. The whole loop — generate → deploy → `DoraEvent` → `devex dora` — was run end-to-end on **real AWS** through OIDC, then torn down.

Key decisions (full ADRs in `docs/architecture/adr/`)

ADR	Decision	Why it matters
0001	Monorepo + self-contained CLI (bundles conventions)	One clone shows the wiring; Git installs resolve reliably
0002	Single DoraEvent emitted by generated CI	DORA comparability + SOC 2 audit are structural, not per-team
0003	Git governance — PR-title convention + ruleset-as-code	Two-reviewer change management is code-reviewable evidence

ADR	Decision	Why it matters
0004	Ship prebuilt dist/ (no build-on-install)	Zero-config, package-manager-agnostic installs (registry-less)

Validated, not just designed

Transactionify (a real, independent Python + CDK service) adopted the golden path with zero application-code changes. Its first standardized CI run **caught six latent bugs** (red → green on one PR), the adoption drove **six platform fix PRs**, and it **deployed to a real AWS sandbox via OIDC**, emitting a real `DoraEvent` that `devex dora` read back to compute the four metrics. See `docs/case-study-transactionify.md`.